

Proof of conjecture 00000002410: the middle-half Cantor measure is not Rajchman

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1 The conjecture and its reading

Statement (English, verbatim). Conjecture: There exists a measure on a $1/2$ -dimensional homogeneous Cantor set whose Fourier transform does not decay to zero (a fractal version of Rajchman counterexamples); the construction uses x_4 -invariant (not x_2) blocking. (The Chinese text says the same.)

Reading. The claim is existential, and we prove it with an explicit witness. This is a classical example; we prove every part of it here. We read the terms as follows.

- *Measure on K :* a Borel probability measure μ on $\mathbb{R}$ with $\mu(K) = 1$.
- *$1/2$ -dimensional:* Hausdorff dimension $\dim_H K = 1/2$ (Mathlib’s `dimH`, for the usual metric on $\mathbb{R}$).
- *Homogeneous Cantor set:* K is a Cantor space, meaning it is homeomorphic to $\{0, 1\}^{\mathbb{N}}$ with the product of discrete topologies [W2]. It is also compact and self-similar for similarities that all have the same ratio. Here $K = (K/4) \cup (K/4 + 3/4)$, with common ratio $1/4$.
- *Fourier transform does not decay to zero:* $\hat{\mu}(\xi) = \int e^{-2\pi i \xi x} d\mu(x)$ does not tend to 0 as $|\xi| \rightarrow \infty$. So μ is not a Rajchman measure, which is a measure “whose Fourier transform vanishes at infinity” [W1]. We also prove the circle version: the Fourier coefficients $\hat{\mu}(m)$, $m \in \mathbb{Z}$, do not tend to 0 as $m \rightarrow \infty$. We also prove the explicit bound $\operatorname{Re} \hat{\mu}(4^n) \geq 1/8$ for every $n \in \mathbb{N}$.
- *x_4 -invariant (not x_2) blocking:* the word “blocking” has no formal meaning. We formalize the invariance it refers to. μ is invariant under $T_4(x) = \operatorname{fract}(4x)$, that is $(T_4)_*\mu = \mu$. It is not invariant under $T_2(x) = \operatorname{fract}(2x)$, that is $(T_2)_*\mu \neq \mu$. Here $\operatorname{fract}(y) = y - \lfloor y \rfloor$. The witness is built from base-4 digits, i.e. from digit blocks of size one in base 4.

Conventions. $\mathbb{N} = \{0, 1, 2, \dots\}$. Digits are indexed from 0: $0.d_0d_1d_2\dots$ in base 4 means $\sum_{i \geq 0} d_i 4^{-(i+1)}$. Any other sign or 2π normalisation of $\hat{\mu}$ only rescales or reflects ξ and conjugates the value, so it does not change the conclusion “ $\hat{\mu} \not\rightarrow 0$ as $|\xi| \rightarrow \infty$ ”. The Lean statement uses the convention displayed above.

2 Definitions (as formalized)

Definition 1. $\operatorname{dig} : \{0, 1\} \rightarrow \{0, \dots, 3\}$ sends $0 \mapsto 0$ and $1 \mapsto 3$. For $a \in \{0, 1\}^{\mathbb{N}}$ let $\operatorname{code}(a) = \sum_i \operatorname{dig}(a_i) 4^{-(i+1)}$. In Lean this is `Real.ofDigits` in base 4, applied to the digit sequence $\operatorname{dig} \circ a$.

Definition 2. $K = \{0.d_0d_1d_2\ldots \text{ (base 4)} : d_i \in \{0,3\} \forall i\}$, the image under `Real.ofDigits` of the base-4 digit sequences with values in $\{0,3\}$ (`cantor`). Lemma `cantor_eq_range` shows that $K = \text{code}(\{0,1\}^{\mathbb{N}})$.

Definition 3. `coin` is the uniform probability measure on $\{0,1\}$ (`PMF.uniformOfFintype`). $\mathbb{P}$ (`bern`) is the product measure `Measure.infinitePi` of countably many copies of `coin`, i.e. the law of i.i.d. fair digits. The Cantor measure is $\mu = \text{code}_*\mathbb{P}$ (`cantorMeasure`). It is the law of $\sum_i d_i 4^{-(i+1)}$ with $d_i \in \{0,3\}$ i.i.d. and uniform.

Definition 4. $\hat{\mu}(\xi) = \int \exp(-2\pi i \xi x) d\mu(x)$ (`fourierTransform`), $T_4(x) = \text{fract}(4x)$, $T_2(x) = \text{fract}(2x)$, and σ is the left shift, $(\sigma a)_i = a_{i+1}$.

3 Proof

Lemma 5 (splitting, gap, separation). (a) For every n , $\text{code}(a) = \sum_{i < n} \text{dig}(a_i) 4^{-(i+1)} + 4^{-n} \text{code}(\sigma^n a)$, and $0 \leq \text{code} \leq 1$. (b) If $a_0 = 0$ then $\text{code}(a) \leq 1/4$; if $a_0 = 1$ then $\text{code}(a) \geq 3/4$. Hence $K \cap (1/4, 3/4) = \emptyset$. (c) If a, a' first differ at index n , then $|\text{code}(a) - \text{code}(a')| \geq 4^{-n}/2$. In particular, `code` is injective.

Proof. (a) is `Mathlib's Real.ofDigits_eq_sum_add_ofDigits`, together with `ofDigits_nonneg` and `ofDigits_le_one`. (b) Take $n = 1$ in (a): $\text{code}(a) = \frac{3}{4}[a_0 = 1] + \frac{1}{4}\text{code}(\sigma a)$. (c) By (a) the two partial sums agree, so the difference is $4^{-n}(\text{code}(\sigma^n a) - \text{code}(\sigma^n a'))$. The two shifted sequences differ at index 0, so by (b) the bracket has absolute value $\geq 1/2$. $\square$

Topology (`cantor_homeomorph`, `isCompact_cantor`, `cantor_selfSimilar`). `code` is continuous (`Real.continuous_ofDigits`) and injective on the compact space $\{0,1\}^{\mathbb{N}}$. So it is a closed embedding, and K is homeomorphic to $\{0,1\}^{\mathbb{N}}$ and compact. By Lemma 5(b), $\text{code}(a)$ is $\text{code}(\sigma a)/4$ or $\text{code}(\sigma a)/4 + 3/4$. Prepending a digit 0 or 1 gives the reverse inclusion, so $K = (K/4) \cup (K/4 + 3/4)$.

Dimension (`dimH_cantor`: $\dim_H K = 1/2$). *Upper bound.* For each n , K is covered by 2^n intervals $[s_w, s_w + 4^{-n}]$, one for each prefix $w \in \{0,1\}^n$ (`cell`), by Lemma 5(a). Then $\sum_w (\text{diam})^{1/2} = 2^n \cdot 2^{-n} = 1$, and `hausdorffMeasure_le_liminf_sum` gives $\mathcal{H}^{1/2}(K) \leq 1 < \infty$. So $\dim_H K \leq 1/2$ (`dimH_le_of_hausdorffMeasure_ne_top`). *Lower bound.* Let $\varphi(\text{code}(a)) = 0.a_0a_1a_2\ldots$ in base 2 (`toBinary`). Let a, a' first differ at n . Then $|\varphi(\text{code } a) - \varphi(\text{code } a')| \leq 2^{-n}$ by `abs_ofDigits_sub_ofDigits_le`, while $|\text{code } a - \text{code } a'| \geq 4^{-n}/2 \geq (2^{-n}/2)^2$ by Lemma 5(c). Hence φ is $1/2$ -Hölder on K with constant 2. Its image contains $[0,1]$ (`Real.ofDigits_SurjOn`), and $\dim_H[0,1] = 1$. So `HolderOnWith.dimH_image_le` gives $1 \leq \dim_H K/(1/2)$.

Measure. μ is a probability measure, and $\mu(K) = \mathbb{P}(\text{code}^{-1}K) = 1$ (`cantorMeasure_cantor`).

Theorem 6 (`re_fourier_ge`). $\text{Re } \hat{\mu}(4^n) \geq 1/8$ for all $n \in \mathbb{N}$.

Proof. By Lemma 5(a), $4^n \text{code}(a) = N + \text{code}(\sigma^n a)$, where $N = \sum_{i < n} \text{dig}(a_i) 4^{n-i-1} \in \mathbb{N}$ (`four_pow_mul_code`). So $\text{Re } \hat{\mu}(4^n) = \int \cos(2\pi \text{code}(\sigma^n a)) d\mathbb{P}(a)$. For $y \in K$ we have $y \in [0, 1/4] \cup [3/4, 1]$, hence $\cos 2\pi y \geq 0$. If moreover the first two digits of y are 0, then $y \leq 1/16$ and $\cos 2\pi y \geq \cos(\pi/3) = 1/2$. The event $E_n = \{a_n = a_{n+1} = 0\}$ has $\mathbb{P}(E_n) = 1/4$, by `Measure.infinitePi_pi`. So the integral is $\geq \frac{1}{2} \cdot \frac{1}{4}$. $\square$

Non-decay. Since $4^n \rightarrow \infty$, the theorem gives $\hat{\mu}(\xi) \not\rightarrow 0$ as $\xi \rightarrow +\infty$ (`not_tendsto_atTop`). Hence $\hat{\mu}(\xi) \not\rightarrow 0$ as $|\xi| \rightarrow \infty$ (`not_rajchman`, filter `cocompact`), and $\hat{\mu}(m) \not\rightarrow 0$ as $m \rightarrow \infty$ in $\mathbb{Z}$ (`not_tendsto_int`).

x4-invariance (`cantorMeasure_T4`). The shift preserves $\mathbb{P}$ (`bern_shift`): its pushforward agrees with $\mathbb{P}$ on boxes, by `Measure.eq_infinitePi`. Next, $4\text{code}(a) = k + \text{code}(\sigma a)$ with $k \in \{0, 3\}$. So $T_4(\text{code } a) = \text{code}(\sigma a)$ unless $\text{code}(\sigma a) = 1$, i.e. unless $\sigma a = (1, 1, \dots)$. That event has probability 0, because singletons are $\mathbb{P}$ -null ($\mathbb{P}\{b\} \leq 2^{-n}$ for all n). Hence $(T_4)_*\mu = \text{code}_*\sigma_*\mathbb{P} = \mu$.

Not x2-invariant (`cantorMeasure_T2_ne`). $\mu(1/4, 3/4) = 0$ by the gap. If $a_0 = 1$ and $a_1 = 0$, then $\text{code}(a) \in [3/4, 13/16]$, so $T_2(\text{code } a) = 2\text{code}(a) - 1 \in [1/2, 5/8] \subset (1/4, 3/4)$. Hence $(T_2)_*\mu(1/4, 3/4) \geq \mathbb{P}\{a_0 = 1, a_1 = 0\} = 1/4 > 0$.

4 Lean formalization and axiom audit

File `lean/Conjecture2410/Basic.lean` (477 lines, only import `Mathlib`, `Lean` and `Mathlib v4.33.1`), namespace `C2410`. The main theorem is:

```
theorem main :
  exists (K : Set R) (mu : Measure R), IsProbabilityMeasure mu /\ mu K = 1 /\
    Nonempty (K homeo (N -> Fin 2)) /\ IsCompact K /\
    K = (fun x => x / 4) '' K U (fun x => x / 4 + 3 / 4) '' K /\
    dimH K = 1 / 2 /\
    (forall n : N, 1 / 8 <= (fourierTransform mu ((4:R) ^ n)).re) /\
    not Tendsto (fourierTransform mu) (cocompact R) (nhds 0) /\
    not Tendsto (fun m : Z => fourierTransform mu m) atTop (nhds 0) /\
    mu.map T4 = mu /\ mu.map T2 != mu
```

(The verbatim block above is an ASCII transcription. In the file, R , N and Z are the blackboard-bold letters and `homeo` is the homeomorphism type.) The witnesses are `cantor` and `cantorMeasure`. Each conjunct is also a named theorem: `cantorMeasure_cantor`, `cantor_homeomorph`, `isCompact_cantor`, `cantor_selfSimilar`, `dimH_cantor`, `re_fourier_ge`, `not_rajchman`, `not_tendsto_int`, `cantorMeasure_T4` and `cantorMeasure_T2_ne`. We use `Mathlib`'s positional-notation library `Mathlib/Analysis/Real/OfDigits.lean` (`Real.ofDigits`, `Real.digits`).

Axiom audit. `#print axioms C2410.main` reports `[propext, Classical.choice, Quot.sound]`. The same holds for `dimH_cantor`, `re_fourier_ge`, `not_rajchman` and `cantorMeasure_T4`. There is no `sorry`, no `native_decide` and no added axiom.

Sources

- [W1] Wikipedia, “Rajchman measure”, https://en.wikipedia.org/wiki/Rajchman_measure: “a Rajchman measure, studied by Rajchman (1928), is a regular Borel measure on a locally compact group such as the circle, whose Fourier transform vanishes at infinity.”
- [W2] Wikipedia, “Cantor space”, https://en.wikipedia.org/wiki/Cantor_space: “a topological space is a Cantor space if it is homeomorphic to the Cantor set.” and “the canonical example of a Cantor space is the countably infinite topological product of the discrete 2-point space $\{0, 1\}$.”